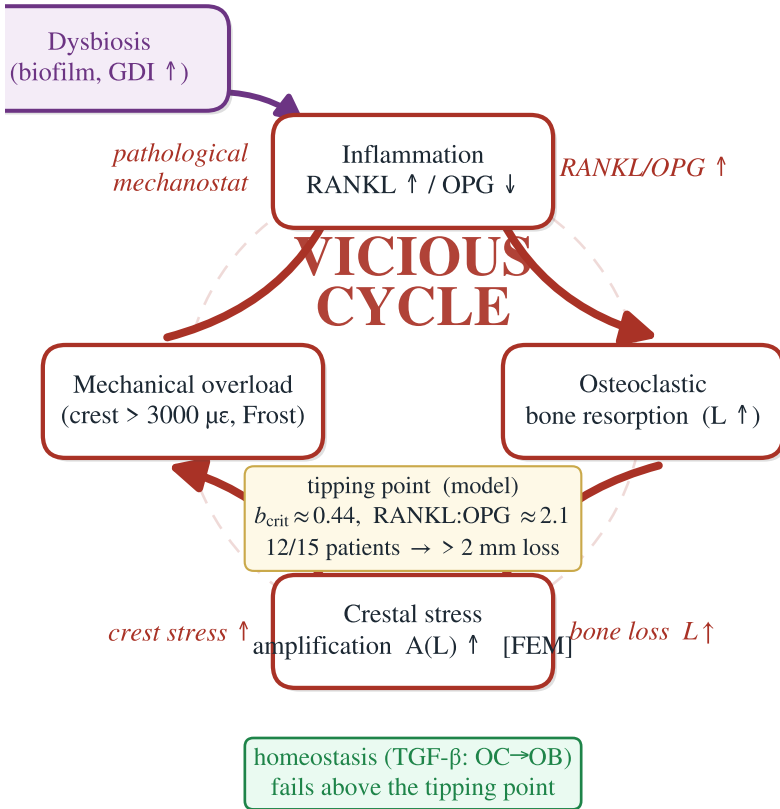


Peri-implantitis mechano-biological vicious cycle



Terms

RANKL / OPG — receptor activator of NF- κ B ligand / osteoprotegerin — their ratio is the master switch for osteoclast (bone-resorbing cell) activation

GDI — guild dysbiosis index, $\log(\text{dysbiotic} / \text{commensal})$ — severity of the microbial imbalance

A(L) — crestal stress-amplification factor as a function of marginal bone loss L (from the FEM bone-loss sweep)

$\mu\epsilon$ (microstrain) — 10^{-6} strain; Frost mechanostat thresholds (>3000 $\mu\epsilon$ = pathological overload → resorption)

TGF- β — growth factor coupling resorption → formation — the homeostatic counter-loop

b_{crit} — dysbiosis level at which the loop tips from arrested to runaway bone loss

Mechanisms: Frost 1987/2003 (mechanostat) · Lemaire 2004, Pivonka 2008, Boyce & Xing 2007 (RANKL/OPG) · Oh 2002 (crestal MBL). Loop closure & A(L) from FEM: this work (model).